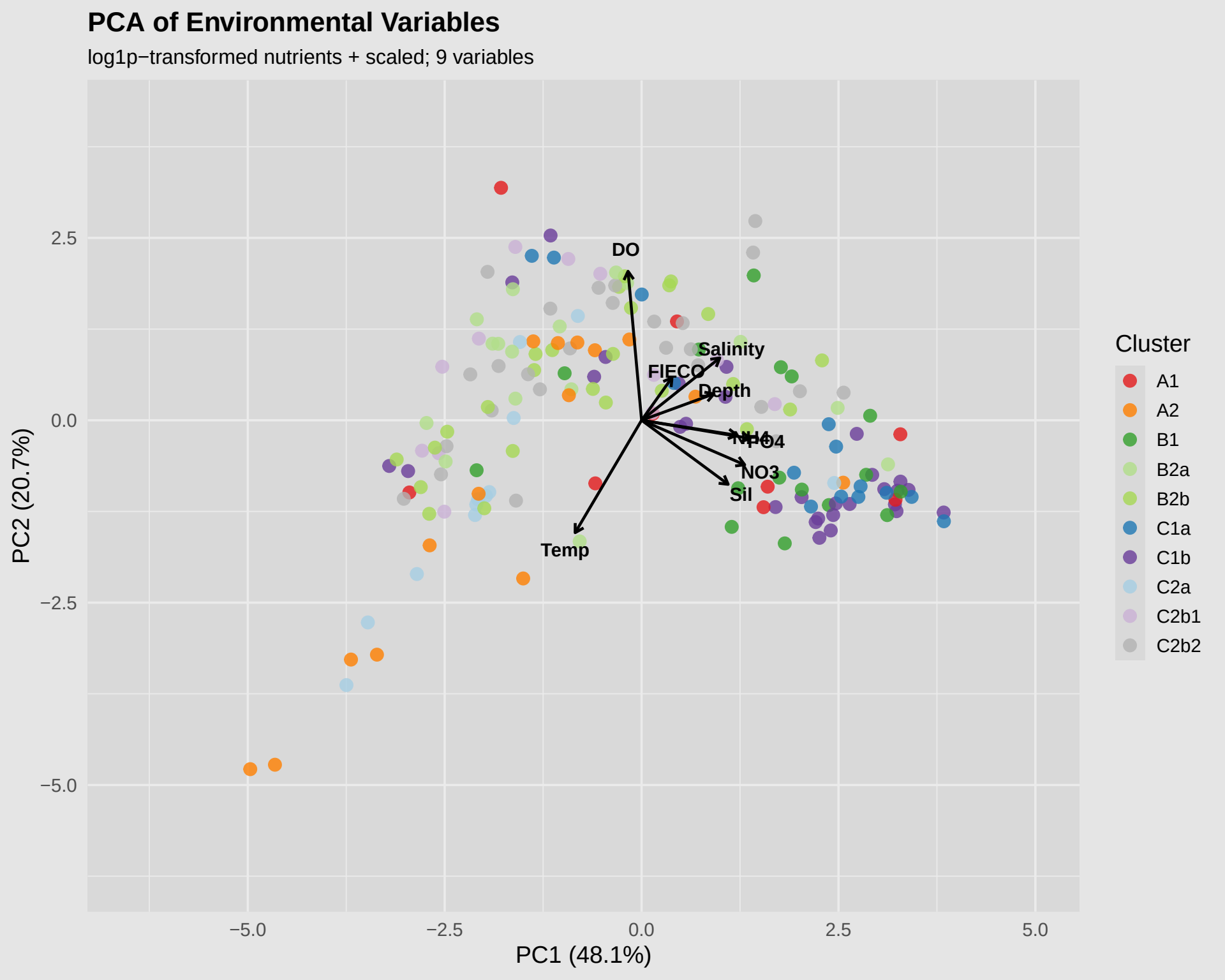
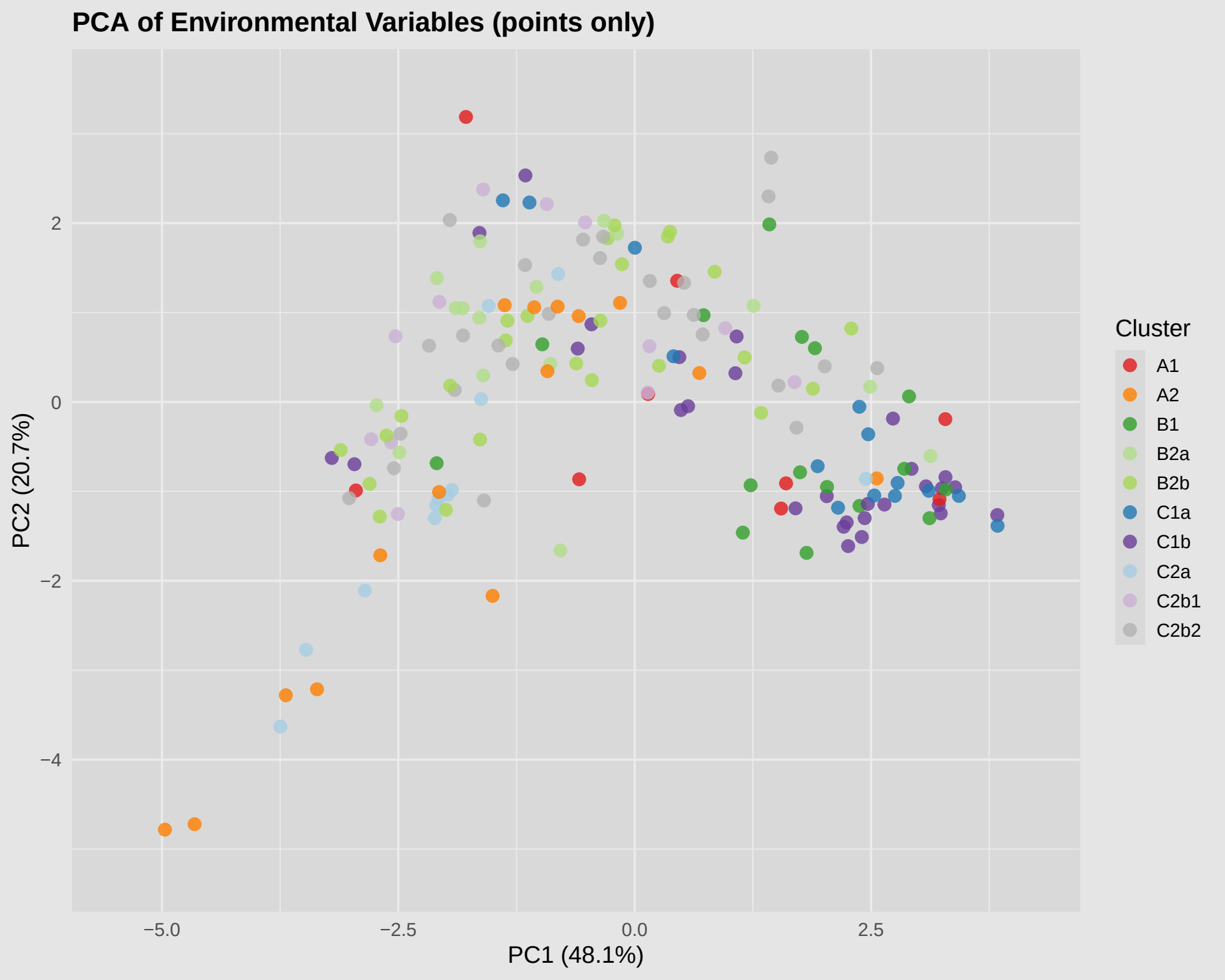
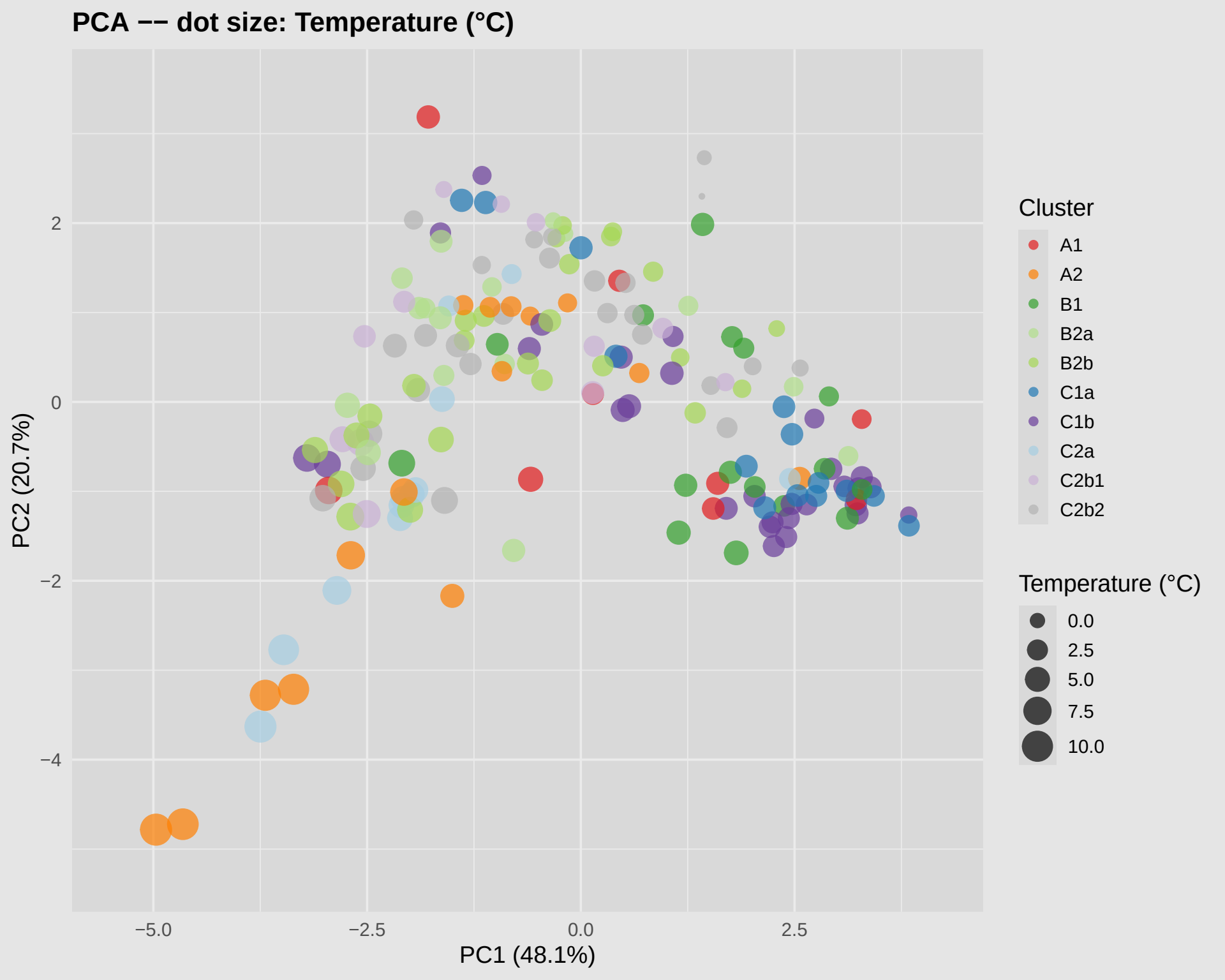
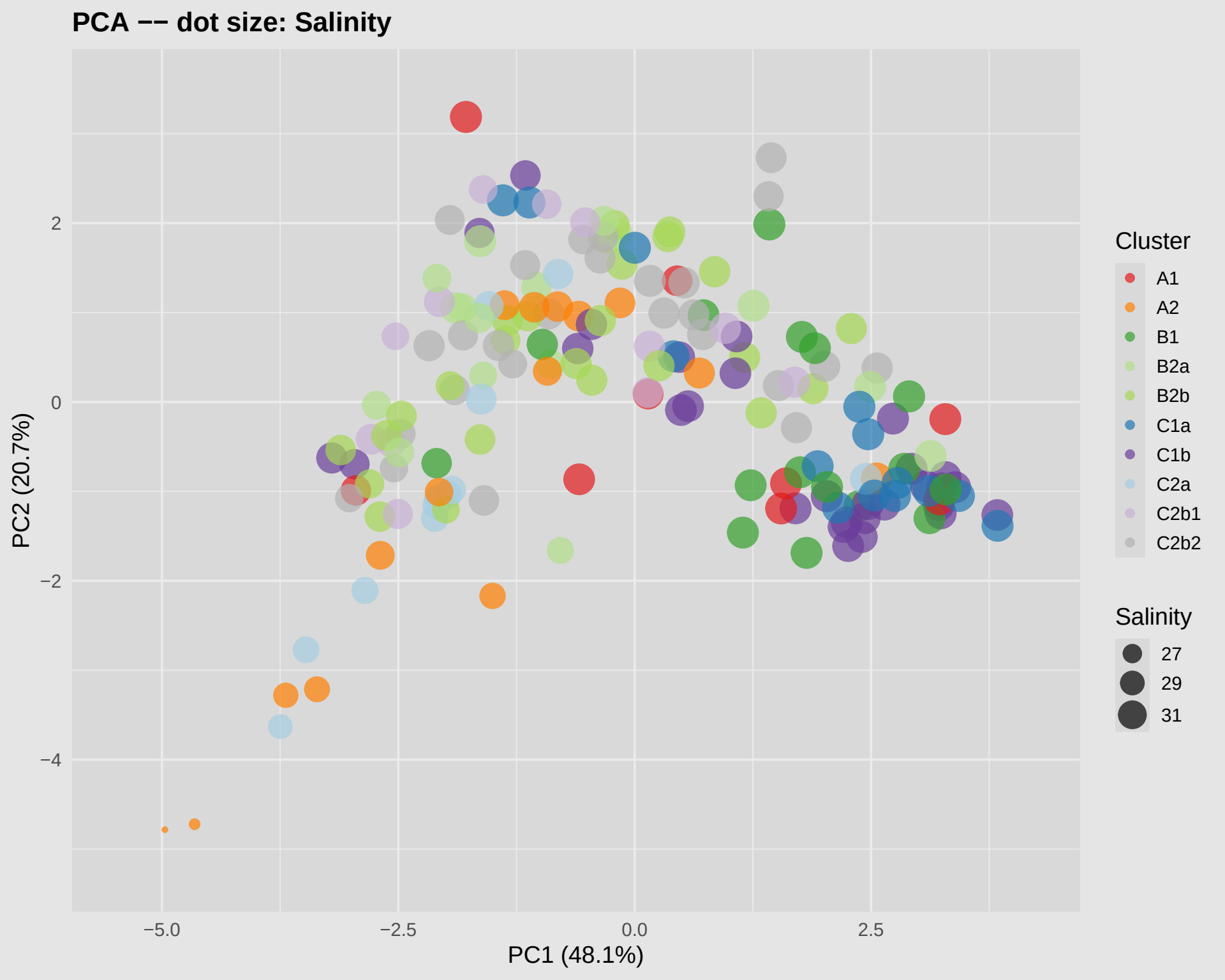
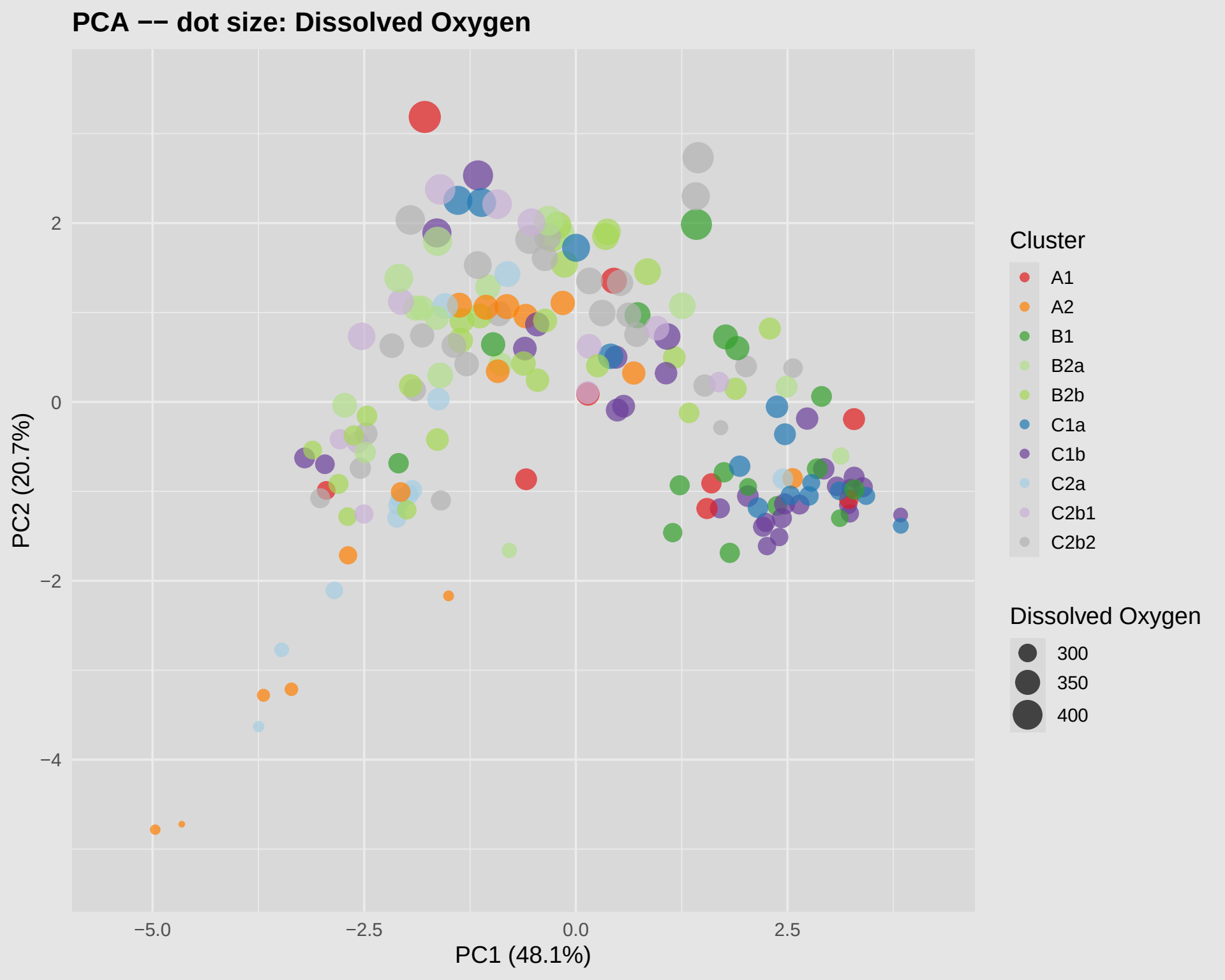
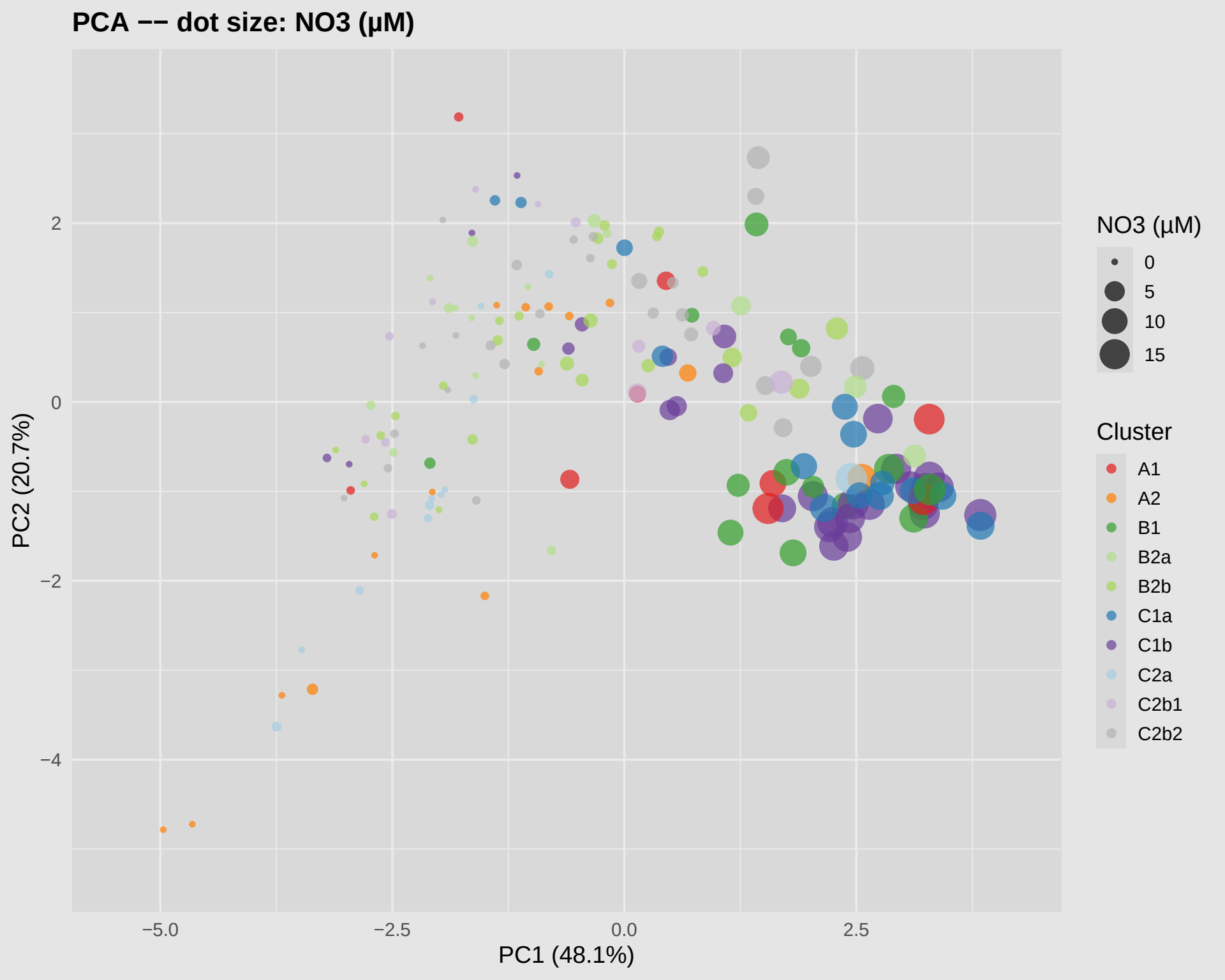

PCA of Environmental Variables
log1p−transformed nutrients + scaled; 9 variables
DO
Salinity
FlECO
Depth
NH4
PO4
NO3
Sil
Temp
2.5
0.0
−2.5
−5.0
PC2 (20.7%)
−5.0
−2.5
0.0
2.5
5.0
PC1 (48.1%)
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2

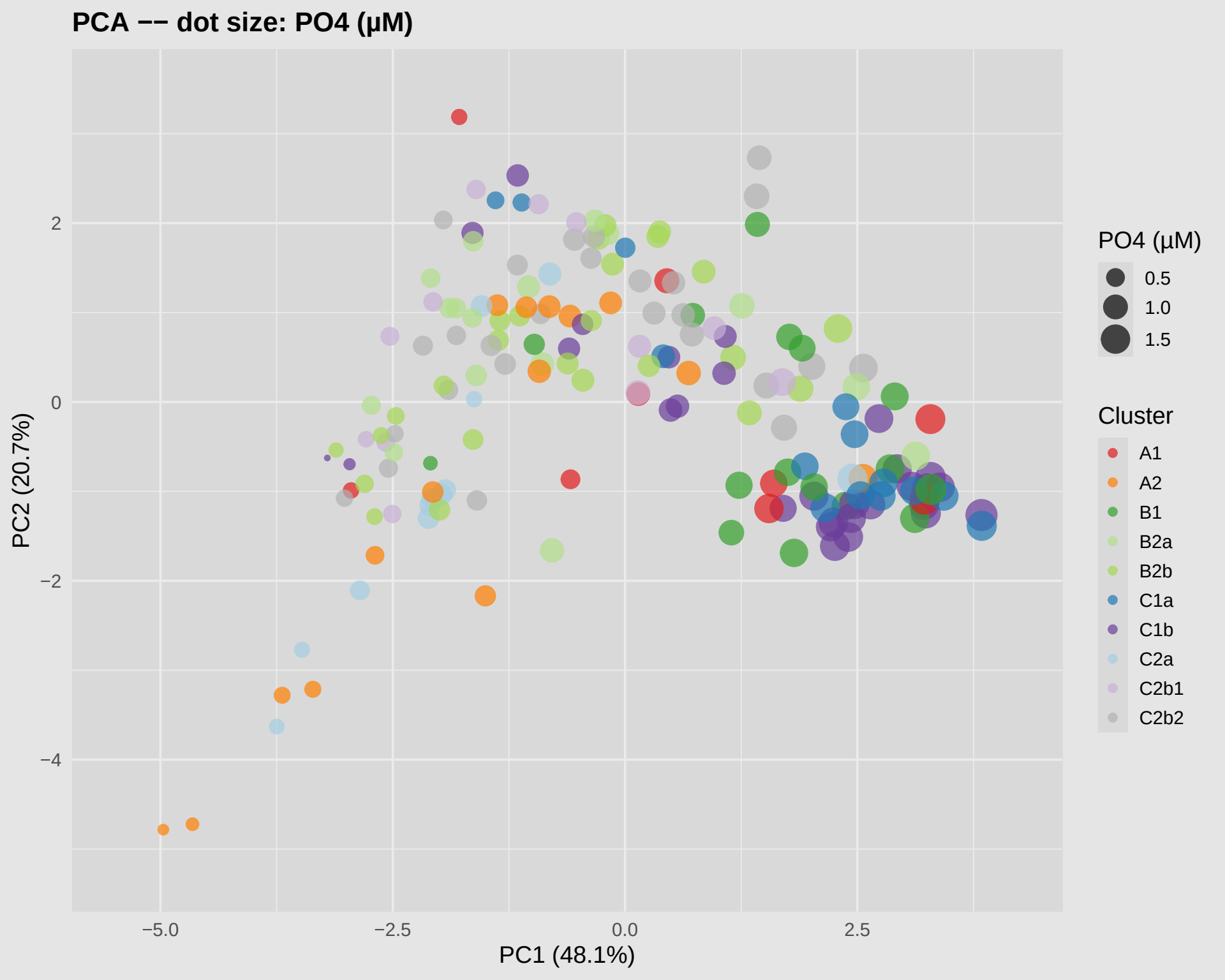

PCA of Environmental Variables (points only)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2

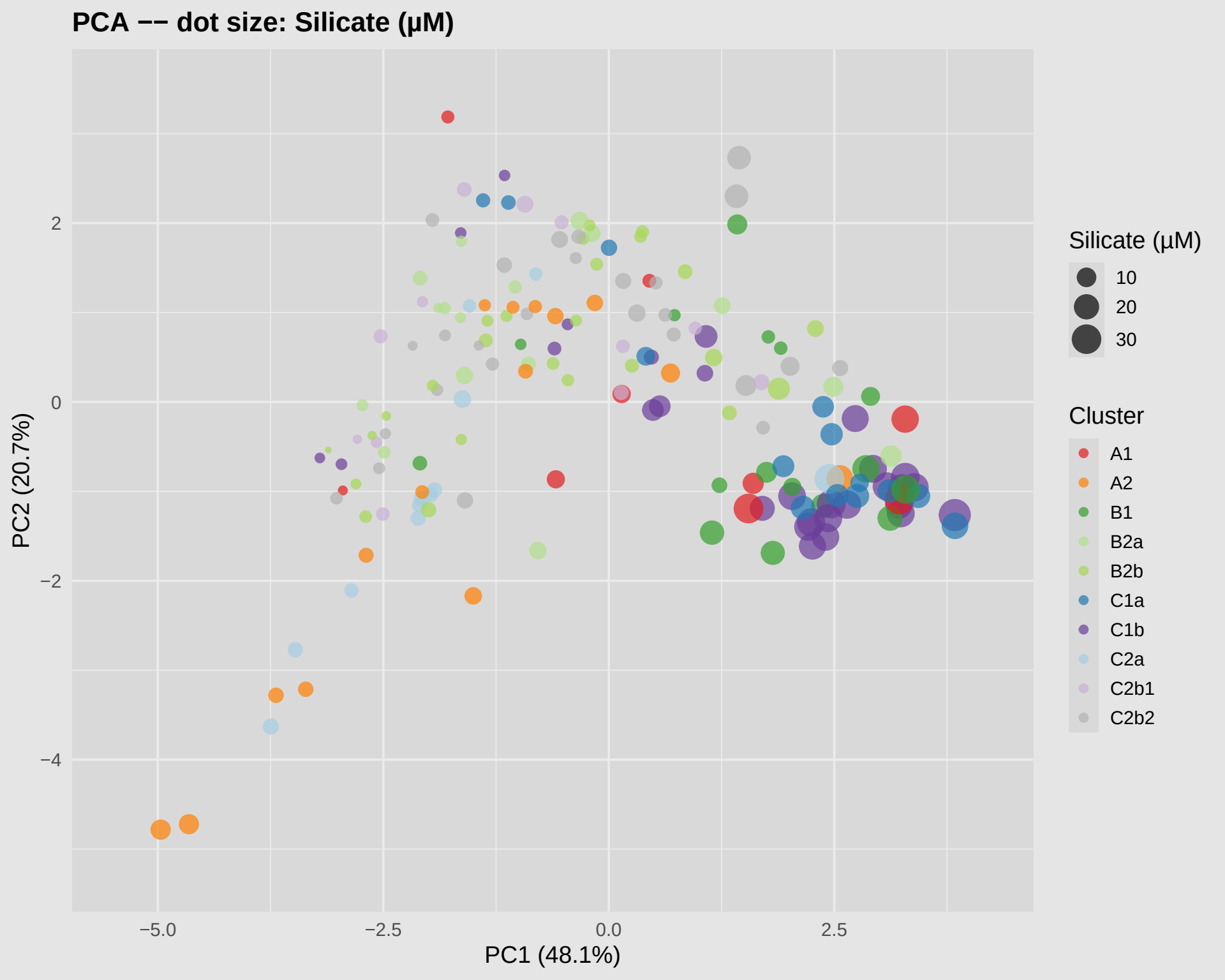

PCA −− dot size: Temperature (°C)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
Temperature (°C)
0.0
2.5
5.0
7.5
10.0

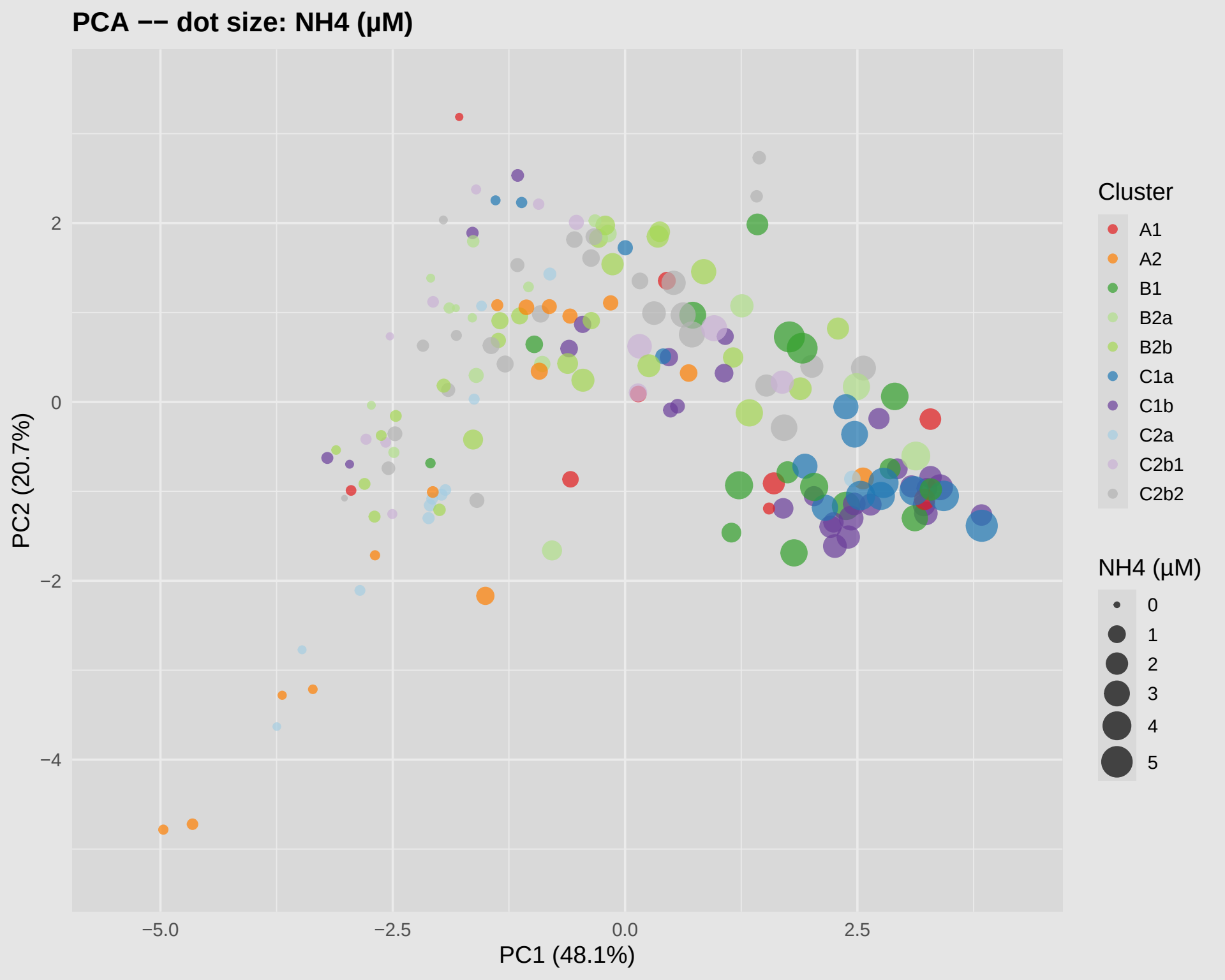

PCA −− dot size: Salinity
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
Salinity
27
29
31

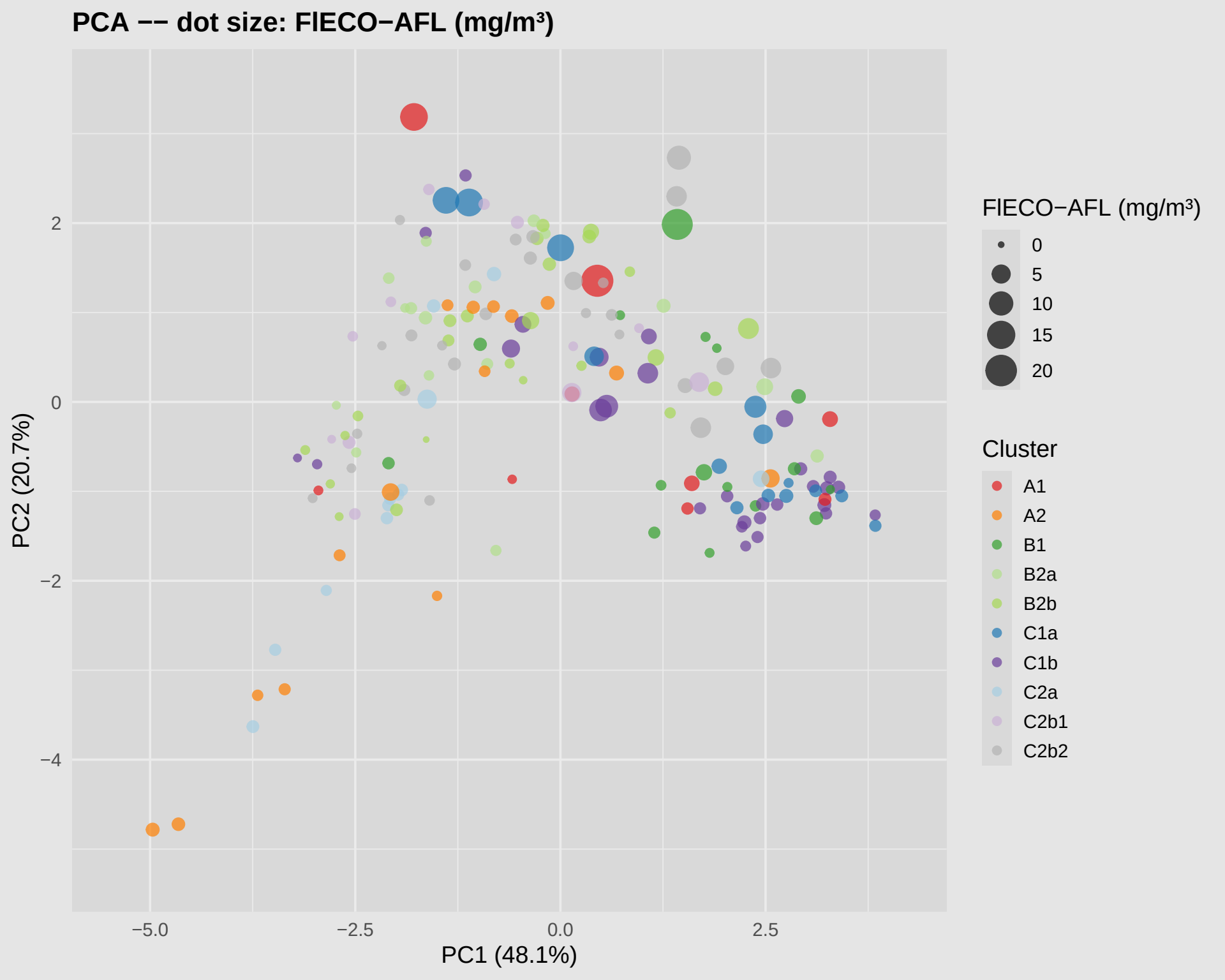

PCA −− dot size: Dissolved Oxygen
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
Dissolved Oxygen
300
350
400

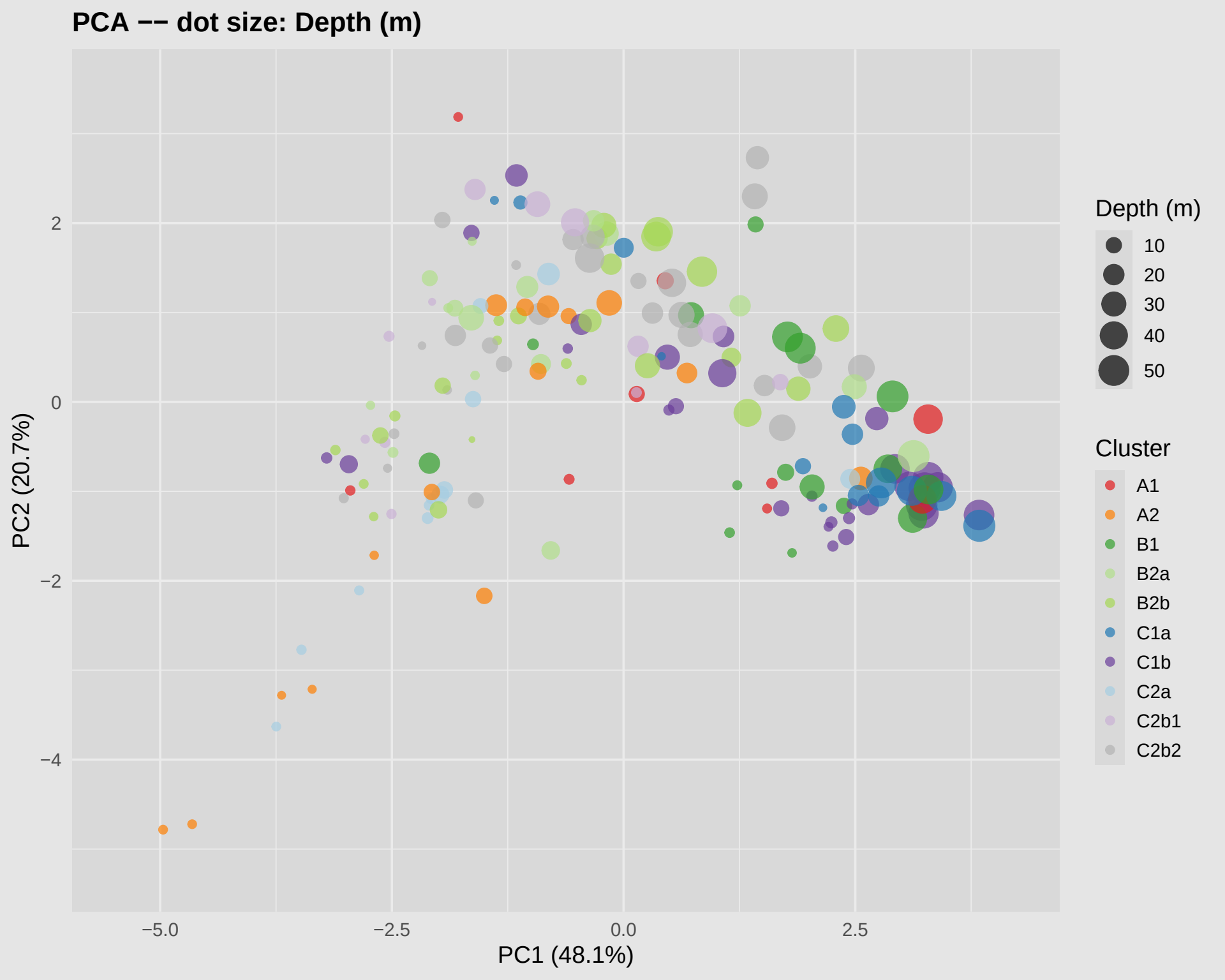

PCA −− dot size: NO3 (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
NO3 (µM)
0
5
10
15
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2

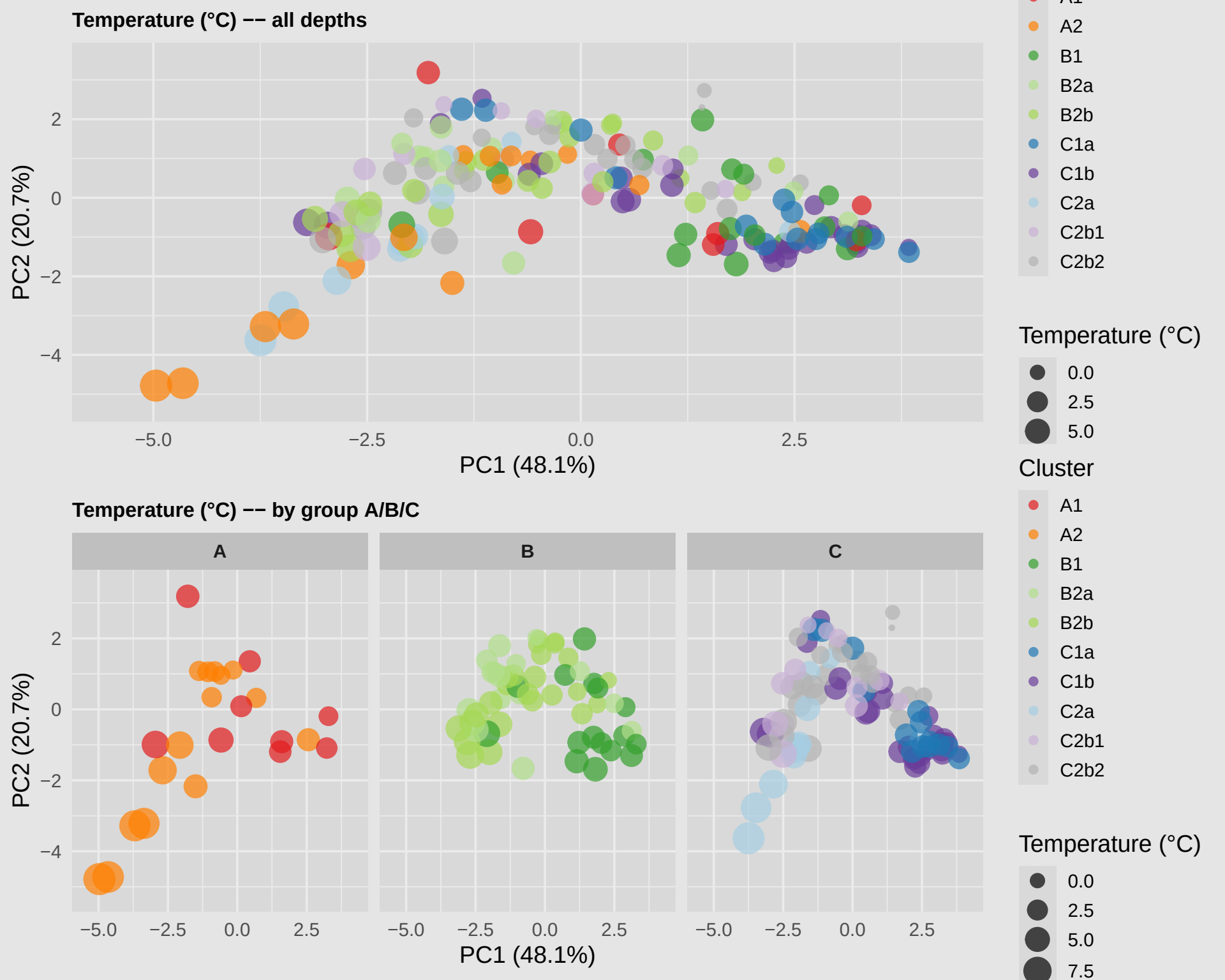

PCA −− dot size: PO4 (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
PO4 (µM)
0.5
1.0
1.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2

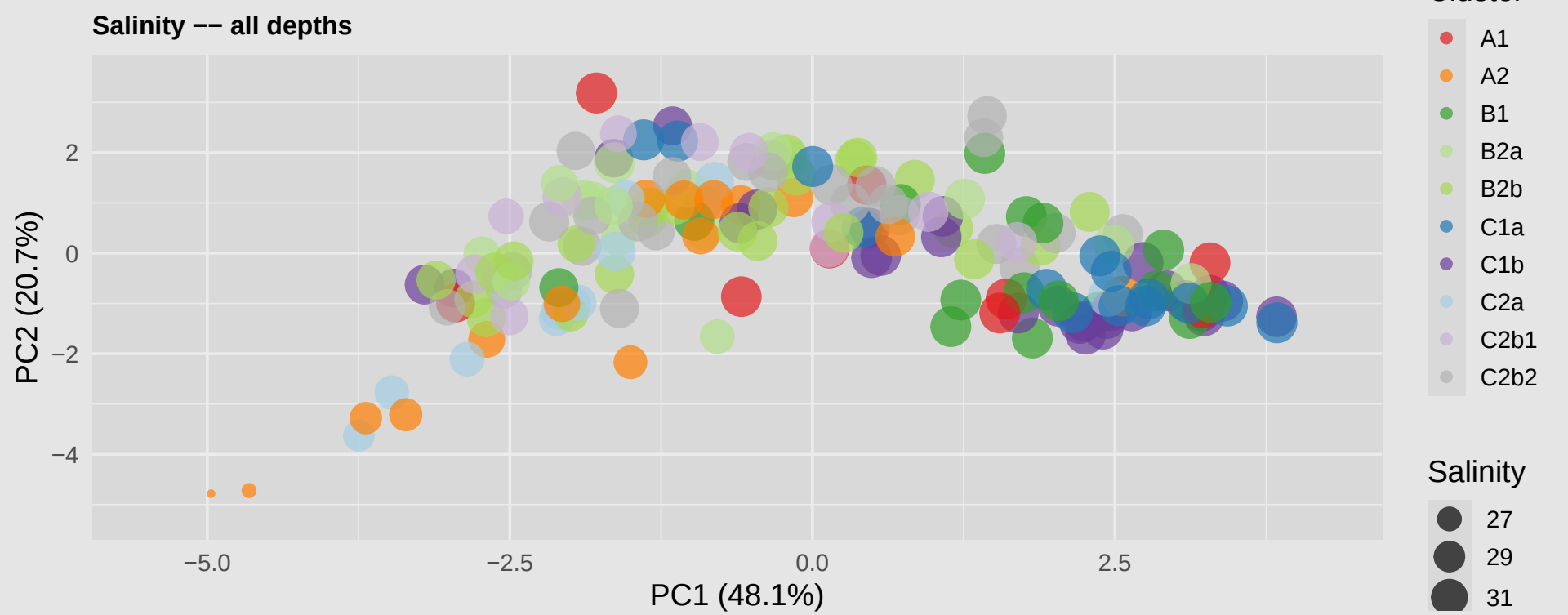

PCA −− dot size: Silicate (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Silicate (µM)
10
20
30
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2

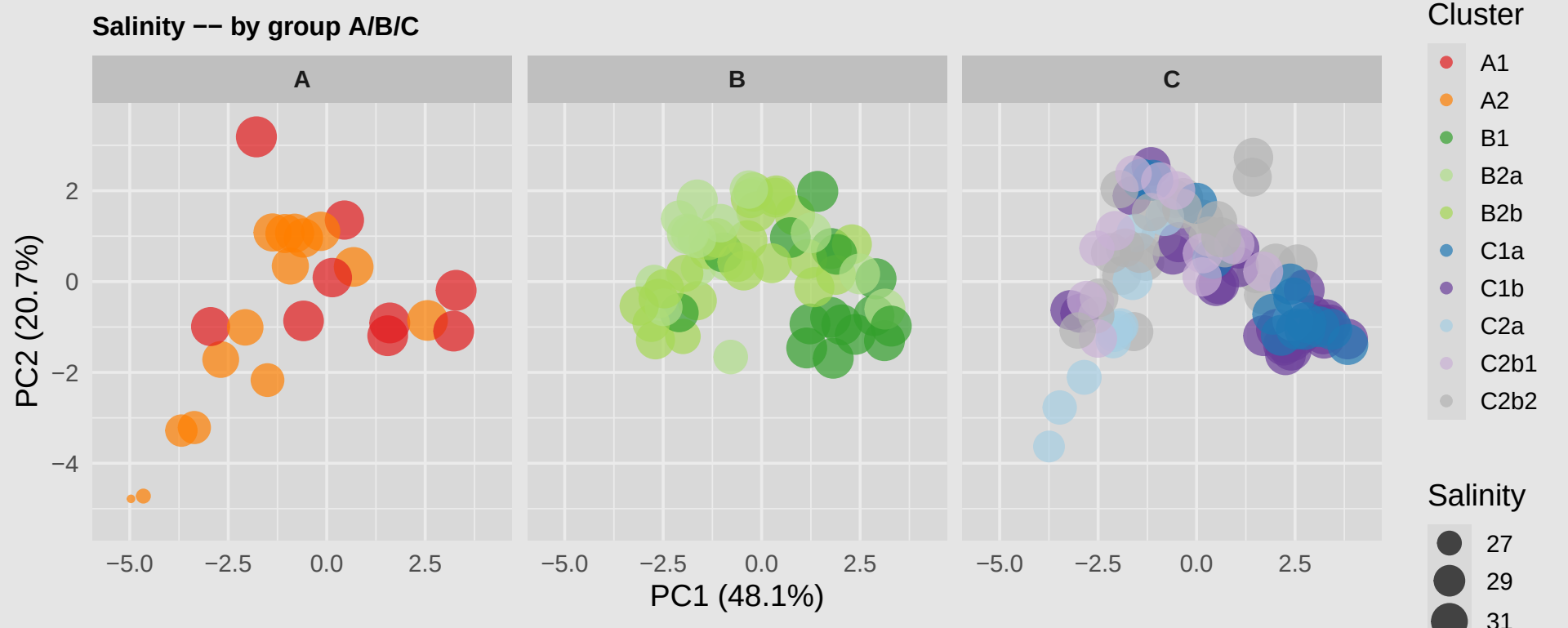

PCA −− dot size: NH4 (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
NH4 (µM)
0
1
2
3
4
5


PCA −− dot size: FlECO−AFL (mg/m³)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
FlECO−AFL (mg/m³)
0
5
10
15
20
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2

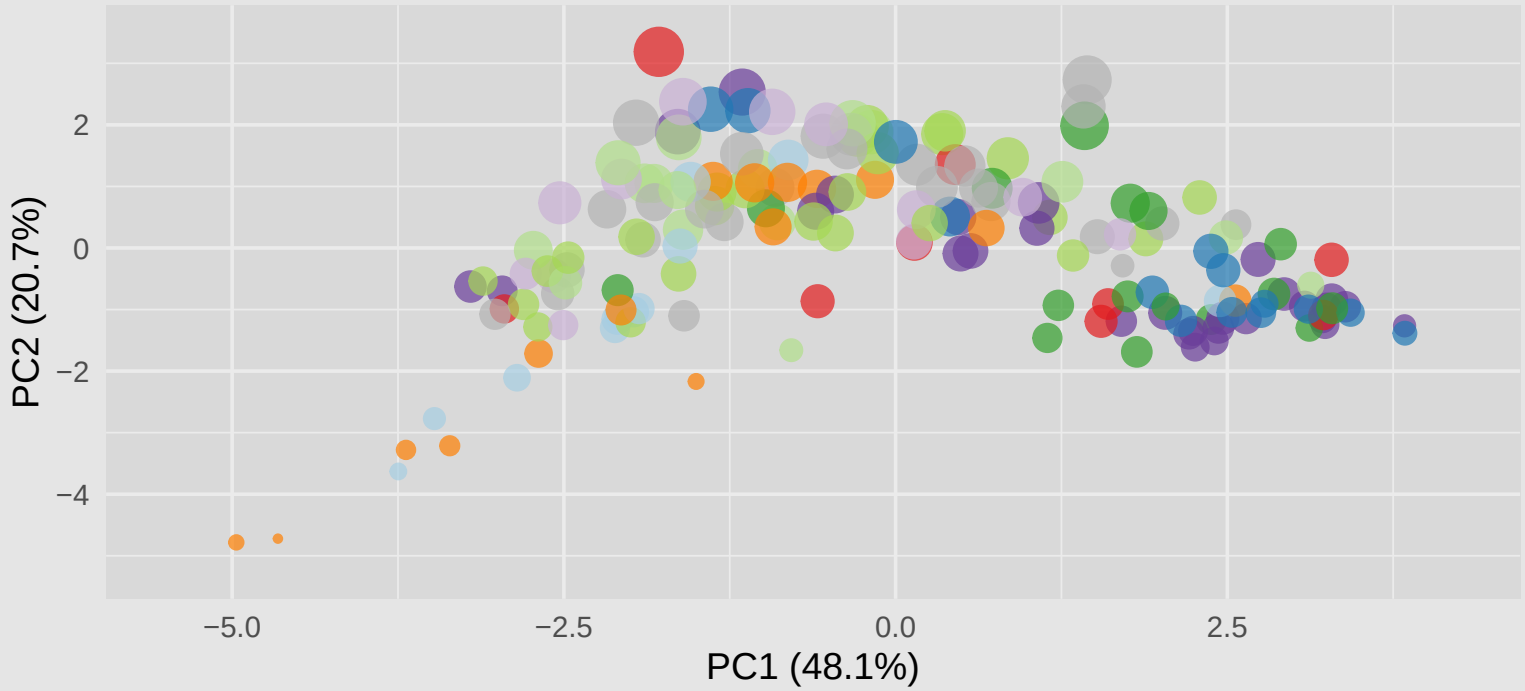

PCA −− dot size: Depth (m)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Depth (m)
10
20
30
40
50
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2


Temperature (°C) −− all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
Temperature (°C)
0.0
2.5
5.0
Temperature (°C) −− by group A/B/C
A
B
C
PC2 (20.7%)
PC1 (48.1%)
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
Temperature (°C)
0.0
2.5
5.0
7.5


Salinity −− all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
Salinity
27
29
31


Salinity −− by group A/B/C
A
B
C
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
Salinity
27
29
31

Dissolved Oxygen -- all depths

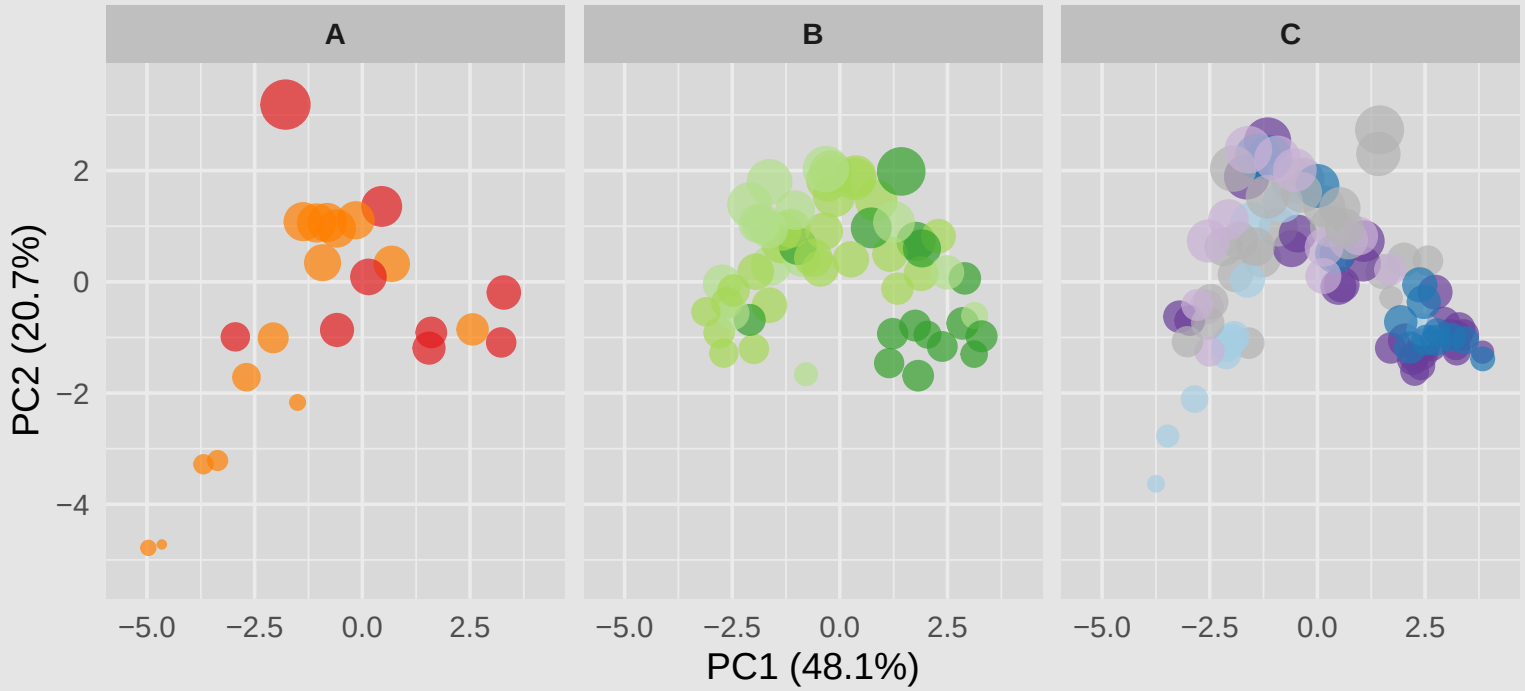


A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2

Dissolved Oxygen

300
350
400

Dissolved Oxygen -- by group A/B/C

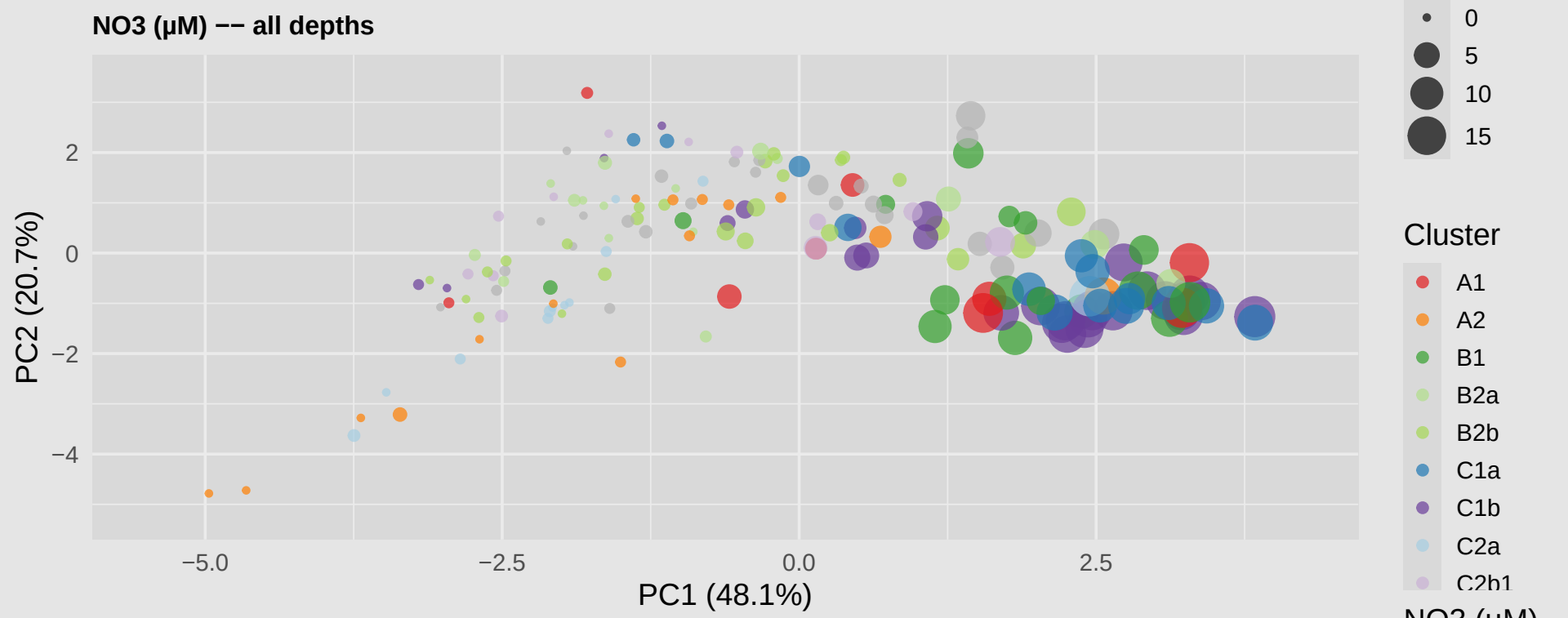


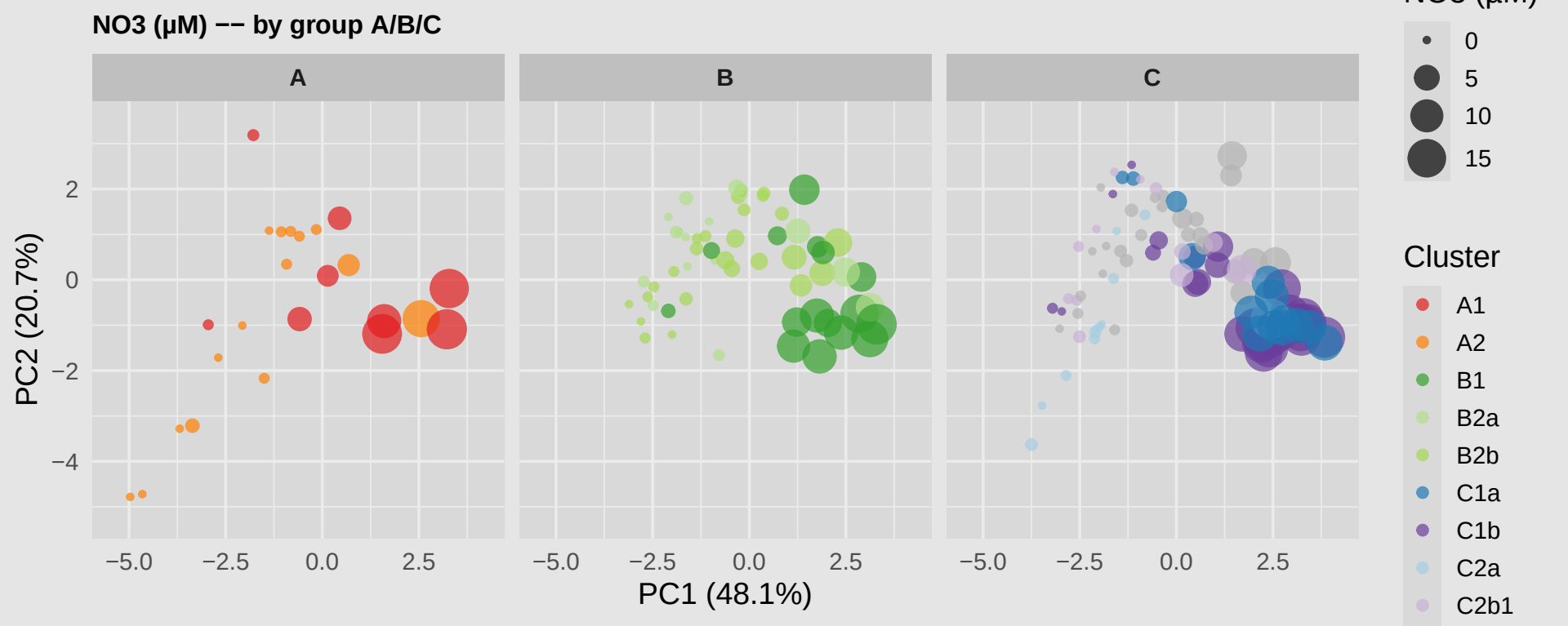
Cluster

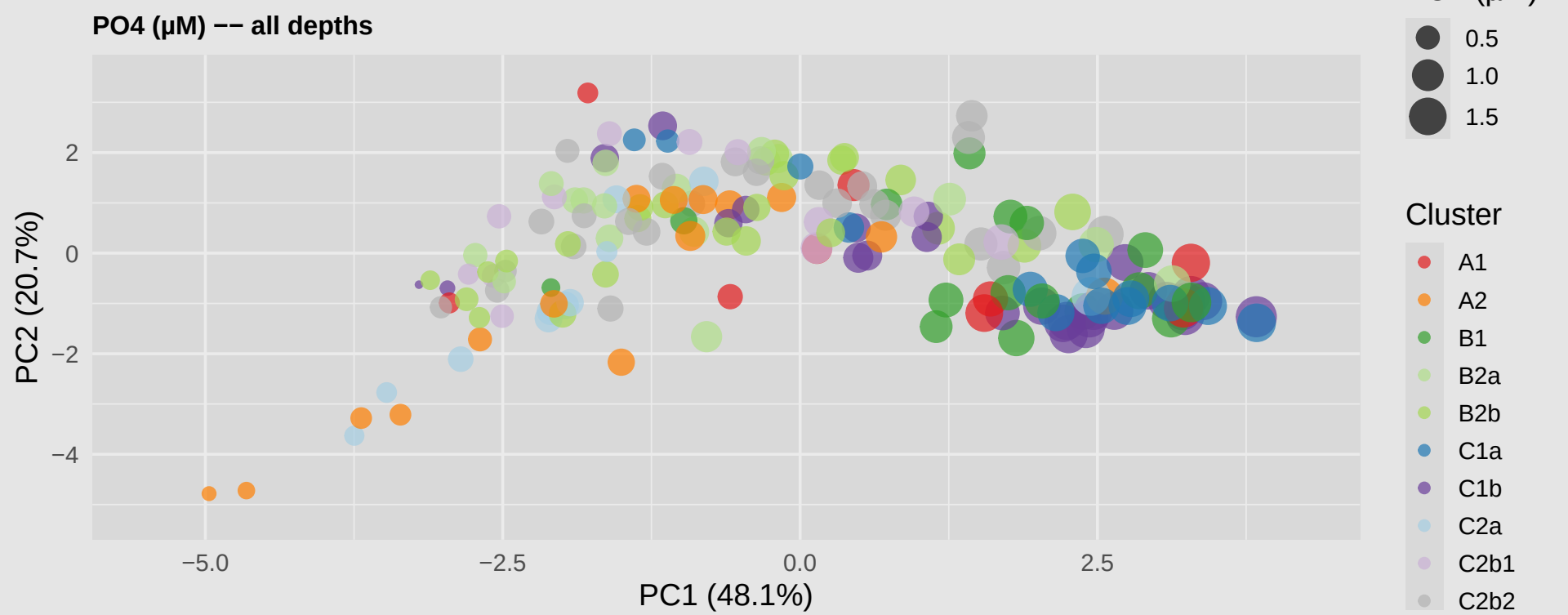
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2

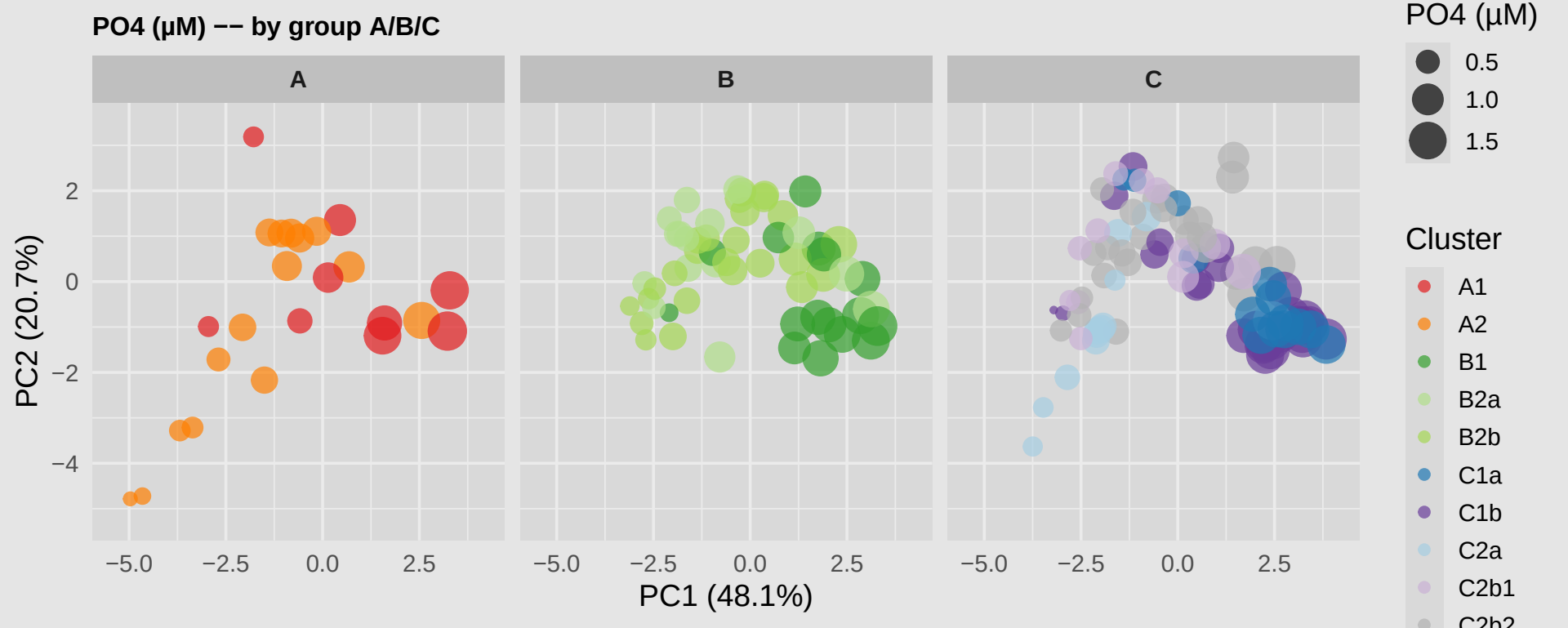
Dissolved Oxygen

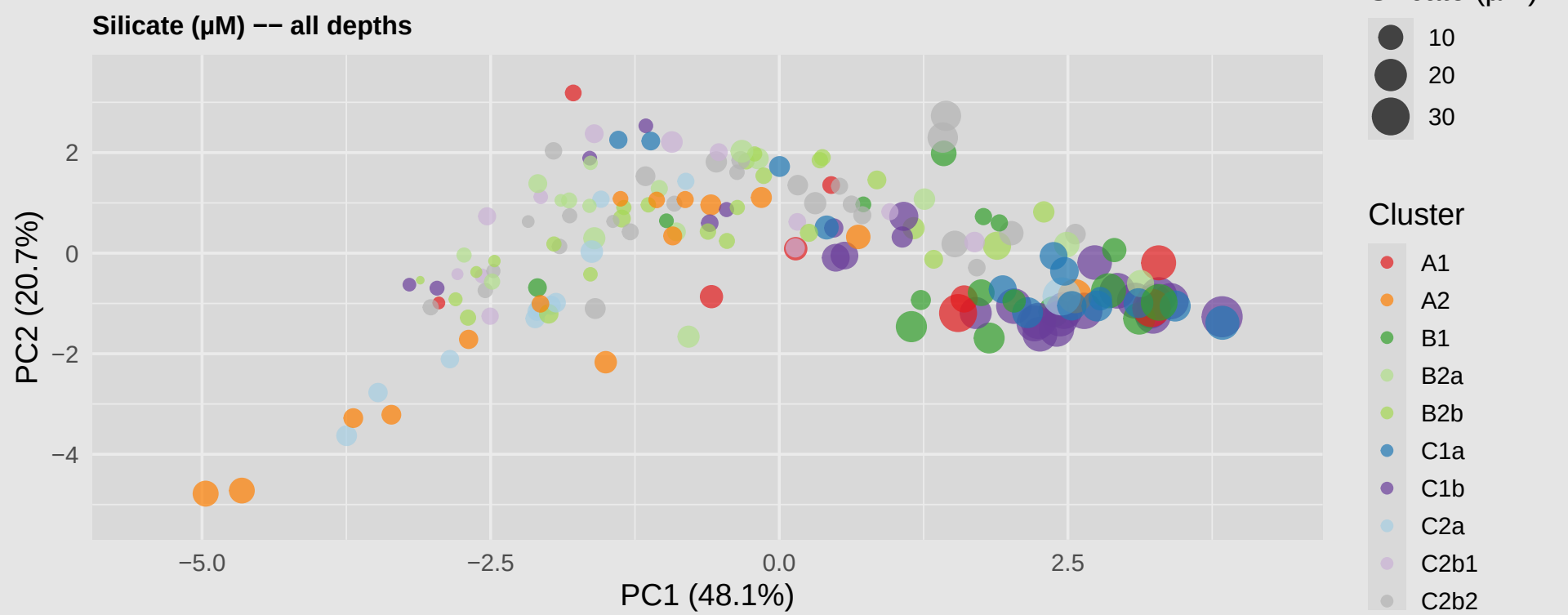
300
350
400

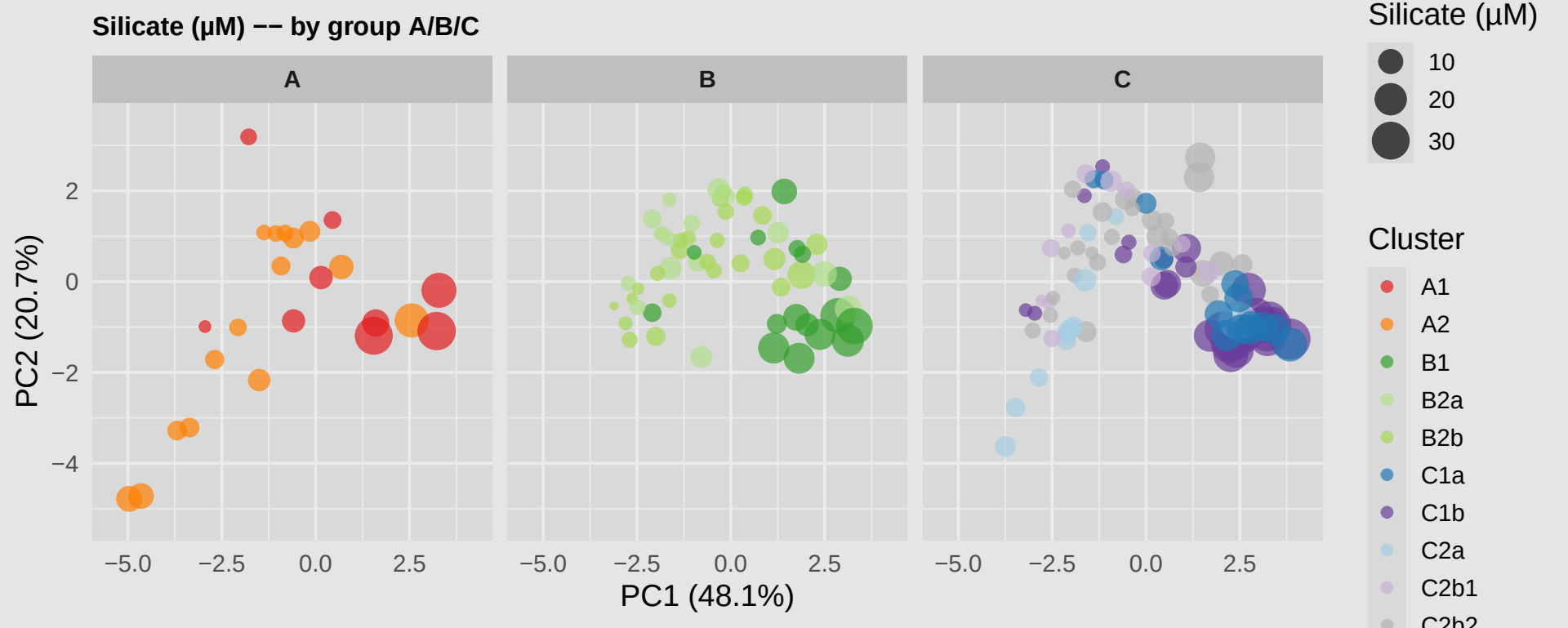

NO3 (µM) −− all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
0
5
10
15
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1


NO3 (µM) −− by group A/B/C
A
B
C
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
NO3 (µM)
0
5
10
15
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1

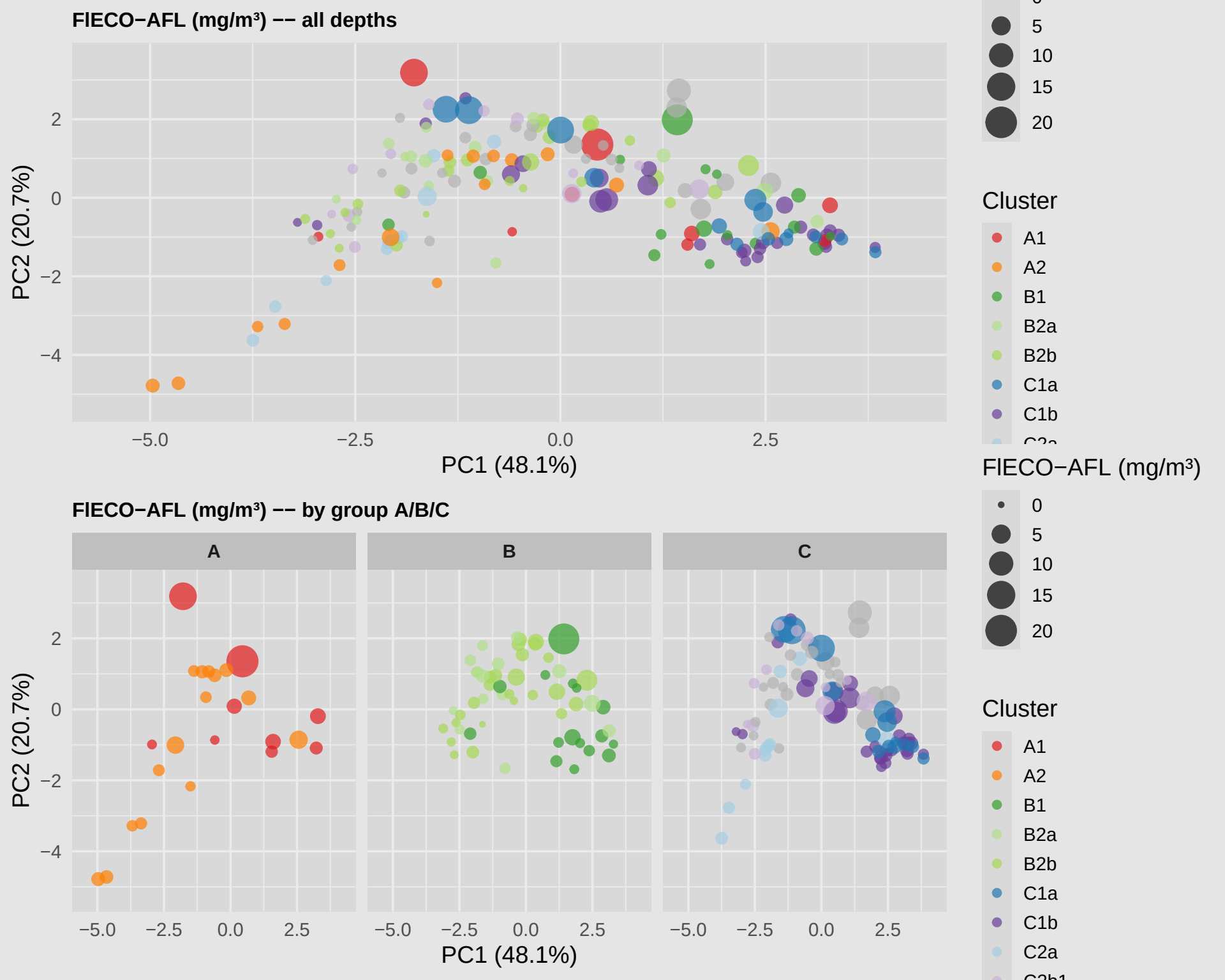

PO4 (µM) −− all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
0.5
1.0
1.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2


PO4 (µM) −− by group A/B/C
A
B
C
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
PO4 (µM)
0.5
1.0
1.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2


Silicate (µM) −− all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
10
20
30
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2


Silicate (µM) −− by group A/B/C
A
B
C
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Silicate (µM)
10
20
30
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2

NH4 (µM) −− all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
NH4 (µM)
0
1
2
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
C2b2
NH4 (µM) −− by group A/B/C
A
B
C
NH4 (µM)
0
1
2
3


FlECO−AFL (mg/m³) −− all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
5
10
15
20
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
FlECO−AFL (mg/m³) −− by group A/B/C
A
B
C
0
5
10
15
20
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a

Depth (m) -- all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
Depth (m)
10
20
30
40
50
Depth (m) -- by group A/B/C
A
B
C
PC2 (20.7%)
PC1 (48.1%)
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b
C2a
C2b1
Depth (m)
10
20
30
40
50